



Gearbox fault diagnosis: A higher order moments approach

Sanjeev Kumar ^{a,b}, Vikash Kumar ^{a,*}, Somnath Sarangi ^a, Om Prakash Singh ^c

^a Department of Mechanical Engineering, Indian Institute of Technology Patna, Bihar, 801103, India

^b Department of Mechanical Engineering, Sikkim Manipal Institute of Technology, Sikkim Manipal University, Sikkim, 737136, India

^c Department of Artificial Intelligence (AI) and Data Science, Sikkim Manipal Institute of Technology, Sikkim Manipal University, Sikkim, 737136, India

ARTICLE INFO

Keywords:

Condition monitoring
Higher order moments (HOM)
Time domain features
ML-classifier
Fault diagnosis

ABSTRACT

The paper presents a higher order moment (HOM) based gearbox fault diagnosis in which the logarithmic amplitude of HOM (LHOM) is proposed as a new feature for gearbox fault diagnosis rather than traditional time domain features. Three benchmark gearboxes datasets at different localised fault conditions are taken to generalise the results obtained from the LHOM approach. The results demonstrate that the LHOM exhibits a linear trend with order of moments and is directly proportional to the severity of faults beyond the fifth order moment. At the end, different case studies are performed by using five popular ML-classifiers and three feature selection techniques on both types of features (conventional and proposed). The results from three distinct gearbox datasets under different localised fault conditions demonstrate that the LHOM features can also be used as extracted features for diagnosing gearbox faults.

1. Introduction

Industry 4.0 places a lot of emphasis on rotating machinery condition monitoring and fault diagnosis. Industrial gearboxes such as spur gearbox [1–3], helical gearbox [2,4], bevel gearbox [5,6], and planetary gearbox [7] are widely used in rotating machinery due to their versatility in power and motion transfer. These gearboxes are practically employed in almost every industry to manage various industrial and precise functionality. However, as a result of repeated loading and functioning under various severe conditions, different types of localised faults develop in the machine component inside the gearbox. These flaws worsen over time, causing anomalous behaviour in the vibration signal, such as impulsive nature at the fault location, non-linearity, and disruption, and its behaviour varies depending on the fault condition. Along with that, machine-generated data or real data contains a lot of noise, which subdues these behaviours inside the signal. Until now, separating these components from the signal for adequate analysis has been a significant barrier for the industry. So, in order to pick out these behaviours and categorise them accurately, various statistical features have been calculated on raw and preprocessed signals in the time domain [5,8–11], frequency domain [11], time–frequency domain [10,12] and are input to the ML-classifiers such as Support Vector Machine (SVM) [13,14], K-Nearest Neighbour (KNN) [8,15], Decision Tree (DT) [6], Random Forest (RF) [3,16], Naive Bayes (NB) [17] and Deep Learning (DL) Algorithm [2,18] for fault diagnostic problem.

Conventionally, most of the time domain statistical features are up to the fourth order moments (namely for bearing and gear damages impulsiveness detection) [19–21] in which the second order moments fully specify the statistics of a Gaussian process. In the case of bearing application, the distribution of vibration signal acquired from healthy and faulty bearing is almost symmetric. Thus its odd central moments are zero and are not useful for bearing diagnostics. Later on, Honarvar et al. [22] showed that the odd order statistics could also be as effective as the even order statistics by rectifying the signal at the initial stage. The findings had been verified by calculating the moments up to 8th order. Unlike the bearing signal, the gearbox signal are more complex, non-Gaussian, and non-linear in nature due to the presence of friction, damping, non-linear stiffness, impulsive peak at localised fault locations, and coupling effects of nearby loaded tooth on gear [23,24]. It is very difficult to specify the statistics of non-Gaussian process with up to 4th order moments, but their higher order statistics carry extra information that can be utilised to measure these deviation from normality [25,26]. To resolve these issues, Martin [27] has used the 6th order and 8th order moments (M6A and M8A) as health indicators for detecting the surface damage on machinery components [28,29], and Decker et al. [30] have concluded that M6A* is sensitive to pick cracked tooth damage of helicopter gearbox.

In literature [10–12], the effect of combined features i.e., time domain clubbed with frequency domain and time–frequency domain features are also studied to improve the overall classification accuracy. These combined features are generally calculated on preprocessed

* Corresponding author.

E-mail address: 1821me18@iitp.ac.in (V. Kumar).

signal obtained through different preprocessor such as wavelet [10], Short-time Fourier transform [12], energy operator [11] to show the effectiveness of these preprocessor in fault diagnosis process. But the major limitations include irrelevant redundant information in raw time domain, selection of proper data-preprocessing tool in pre-processed time domain, time domain information loss in frequency domain, dependency to choose correct wavelet and wavelet scale in time–frequency domain, selection of optimal parameters in phase-space reconstruction domain and so on [31].

In recent years, higher-order statistics (HOS) signal processing is extensively used to a wide range of science and engineering fields, including communications, sonar, geophysics, image processing, biomedicine, radar, oceanography, speech processing, plasma physics, and fluid dynamics [32]. Its powerful qualities stem from the enormous quantity of information carried by higher order moments (HOM) of signals compared to the limited information carried by their second-order statistics help to cope the limitations. As a result, knowing the HOM of non-Gaussian processes, such as many real-world signals, might be beneficial. Furthermore, HOM domains increase the signal-to-noise ratio under wide range of signal and noise situations, allowing for better detection, parameter estimation, and signal reconstruction [32,33].

1.1. Motivations and contributions

Recent years have seen extensive research on gearbox condition monitoring and defect diagnostics. The statistical features extracted from the gearbox's signal in different domain play a major role in diagnosing the gearbox faults. Instead of time domain analysis, other domain need extra preprocessor for improving the fault diagnosis technique. Till now, the statistical indicators up to the fourth order moment, as well as the sixth and eighth order moments, have been used for detecting particular or specific gear faults. It is to be noted that 6th and 8th order moments were examined for gearbox fault detection (two class problem) rather than the multi-class gearbox fault classification problem. Mostly the extracted features in time domain (combination of up to 4th order moment) are fed to the ML-classifier for fault diagnosis problem. Therefore, motivated by HOS signal processing for real-world non-Gaussian signals, the paper proposes HOMs-based feature extraction for gearbox fault diagnosis. For the first time, this paper presents a new feature based on the logarithmic amplitude of higher-order moments (LHOM) for gearbox fault diagnosis rather than traditional time-domain features. This study is carried out on three benchmark datasets taken from different gearboxes (such as spur, helical and bevel) under different localised gear tooth fault conditions (such as cracked-small; medium; large, chipped, broken, missing) to generalise or validate the results obtained from proposed LHOM feature approach. First, the LHOM features at different localised faults condition are calculated and plotted with respect to order of moments at different operating parameters like constant speed, varying speed, no load, high load to predict the behaviour of LHOM features under such conditions. Furthermore, its effectiveness as extracted features are checked with different well-known ML-Classifiers such as KNN, Multi-Class SVM (MCSVM), DT, RF, and NB at three feature selection technique i.e., Pearson Correlation Coefficient (PCC), Neighbourhood Component Analysis (NCA), and Relief-F (RF). Fig. 1 shows the flow chart for gearbox fault diagnosis using proposed LHOM features with NCA and MCSVM. At the end, a comparative analysis between the both types of features (conventional and proposed) are discussed.

The remaining section of the paper is organised as follows: Section 2 discusses the Feature Extraction, Selection and Classification Techniques. Section 3 describes the three benchmark datasets for diagnosing gearbox faults. Section 4 describes the results and important findings from this research work. The paper is concluded in Section 5, which also discusses some future aspects of finding of this research work.

2. Feature extraction, selection, and classifiers

In this section, we will be discussing about the proposed LHOM feature extraction technique, three different feature selection techniques (PCC, NCA, Relief-F) and popularly used ML-Classifiers such as MCSVM, KNN, DT, NB, RF.

2.1. Log amplitude of HOM (LHOM) feature extraction

The standardised moment or the normalised central moment i.e., M_k of order k is given by [32,33]:

$$M_k = \frac{\frac{1}{m} \sum_{i=1}^m (x_i - \mu)^k}{\sigma^k} \quad (1)$$

In this Eq. (1), M_k is normalised by standard deviation (σ) of same order as that of the moment. x_i is the i th datapoint in the sample, m is the total number of datapoints in the sample, $k = [1, 2, 3, \dots, 100]$ is the order of moment (vary up to 100 in this work), and μ is the mean of m datapoints in the sample. These are dimensionless values that represent the signal sample distribution irrespective of their linear change of scale. These indicate the different shape features of the signal. HOM (moments beyond 4th-order) are used for the estimation of further shape parameters. HOM, unlike second-order moments, are not phase blind since they are asymptotically unaffected by additive Gaussian disturbances and the phase information is not fully lost inside the signal. HOM give additional subtle information of the distribution. With the help of HOM, probability density function can be easily calculated. The HOM show extreme variation and carry the peakedness information along with the asymmetry of the signal. As a result, the amplitude of HOM can be used to distinguish the different health conditions of the system. Due to its four basic properties i.e., ability to suppress the additive Gaussian noise with uncertain power spectrum; ability to reconstruct non-minimal phase signals; ability to extract information from Gaussianity deviations; ability to detect and characterise non-linear features in the signals, the HOM can be used for feature extraction for fault diagnosis problem.

In this work, the formulation of M_k is modified for initial orders ($k = 1, 2$) because the central moment of 1st order is zero and standardised central moment of 2nd order becomes 1. So, in order to make them effective, the modified M_k , i.e., $M1_k$ is calculated as follows:

$$M1_k = \begin{cases} \frac{1}{m} \sum_{i=1}^m (x_i), & k = 1 \quad (\text{Non standardised Non central moment}) \\ \frac{1}{m} \sum_{i=1}^m (x_i - \mu)^k, & k = 2 \quad (\text{Non standardized Central moment}) \\ \frac{\frac{1}{m} \sum_{i=1}^m (x_i - \mu)^k}{\sigma^k}, & k > 2 \quad (\text{Standardised Central moment}) \end{cases} \quad (2)$$

Further, the log amplitude of $M1_k$ i.e., LHOM is calculated so that the value of $M1_k$ could be comparable and effective for fault diagnosis problem, and is given by:

$$\text{LHOM} = \log_{10} |M1_k| \quad (3)$$

These LHOM features are proposed in this work as new features for fault diagnosis problems.

2.2. Feature selection techniques

2.2.1. Pearson Correlation Coefficient (PCC)

The PCC feature selection technique is employed in this paper for ranking the features and it can be defined as [34]:

$$P_i = \frac{\sum_i (a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\sum_i (a_i - \bar{a})^2 (b_i - \bar{b})^2}} \quad (4)$$

Where a_i is the i th feature and b_i is its corresponding label. Also, $\bar{a}$ and $\bar{b}$ are its group mean. The absolute value of the correlation coefficient

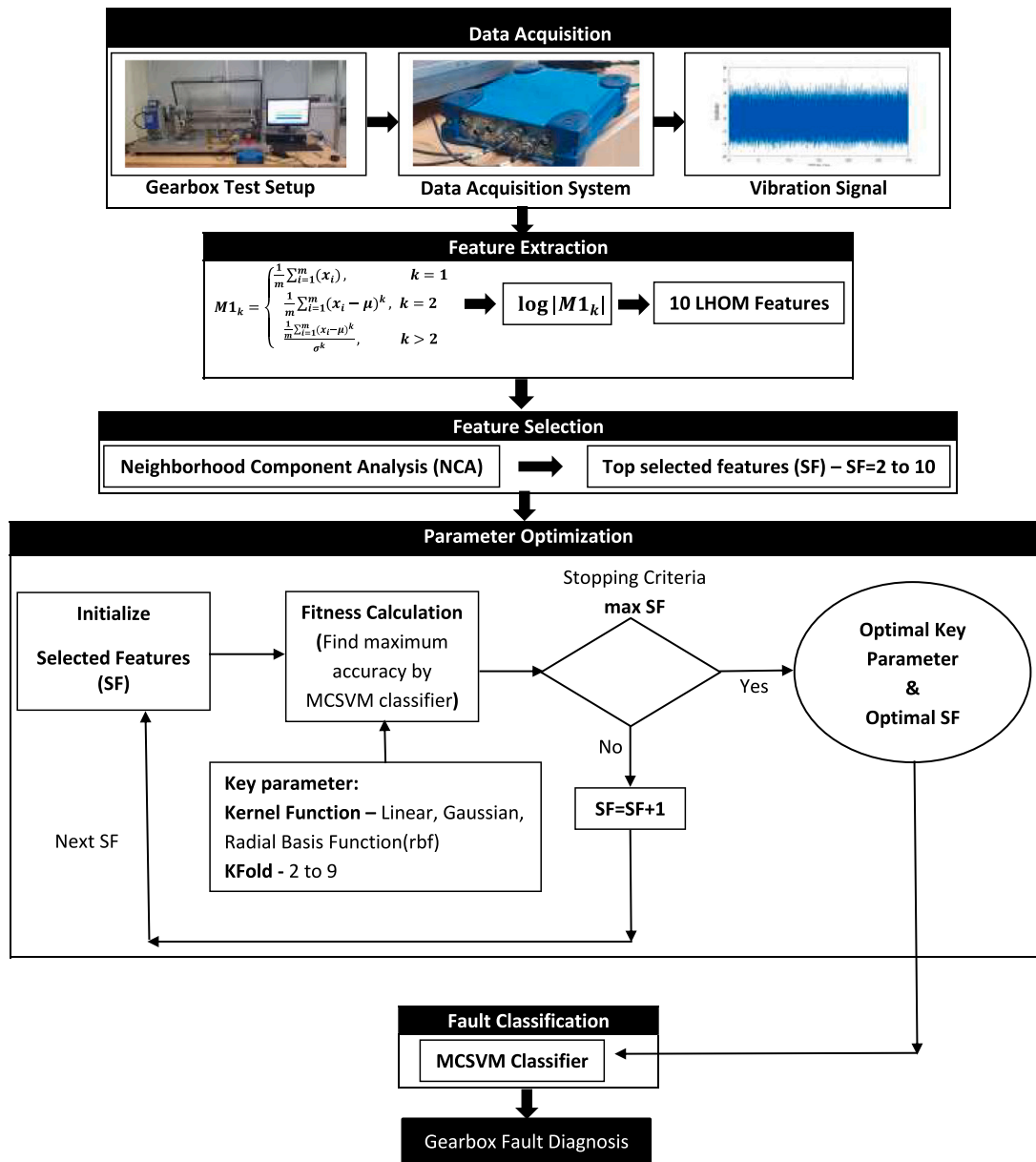


Fig. 1. Flow chart for gearbox fault diagnosis using proposed LHOM features with MCSVM Classifier.

P_i is employed to quantify the usefulness of a trait in separating both classes, with larger values can easily separate the classes. Finally, features that exceed a user-defined threshold are accepted for the analysis. The threshold values are fixed based on the Cross-validation procedure and minimum generalisation error. The PCC are employed in various literature to estimate the significant features. This feature selection technique is best in finding the linear correlations between features and targets [34].

2.2.2. Neighbourhood Component Analysis (NCA)

The purpose of neighbourhood component analysis (NCA) [35] is to maximise the prediction accuracy of regression and classification algorithms. NCA feature selection with regularisation is used to learn feature weights for the purpose of minimising an objective function that estimates the average leave-one-out classification or regression loss over the training dataset $S(x_i, y_i)$, $i = 1, 2, \dots, n$. This method is similar

to a 1-NN classifier in that the reference point is chosen to be the nearest neighbour of the new point x . The reference point in NCA is picked at random, and all points in training set have some chance of being chosen as the reference point.

2.2.3. Relief-F (RF)

The Relief-F feature selection algorithm [36] is modification of the original Relief algorithm [37] which was limited to two-class classification problem. Relief-F finds the weights of predictors in the case where y is a multiclass categorical variable. Predictors who give different values to neighbours of the same class are penalised, whereas predictors who give different values to neighbours of different classes are rewarded. First, Relief-F sets all prediction weights W_j to 0. The programme then iteratively selects a random observation x_r , finds the k -nearest observations to x_r for each class, and updates all the weights for the predictors F_j for each nearest neighbour x_q .

2.3. Classifiers

2.3.1. Multi-Class Support Vector Machine (MCSVM)

Support Vector Machine (SVM) is a supervised machine learning technique. It aims to discover the best border (hyperplane) between multiple classes. In simple terms, SVM performs sophisticated data modifications based on the specific kernel function, with the goal of maximising the separation boundaries between data points. It is a predictive classification and regression approach that use supervised machine learning to improve the classifier's predicted performance. For multi-class problem, two models i.e one-vs-all and one-vs-one are widely used. In one-vs-all model, N separate binary models are created to classify N class problem in which the data associated to N th model assign +ve class and the rest assign -ve class while in one-vs-one model, $\frac{N(N-1)}{2}$ binary models are created in which each model is trained for two class at a time and step repeats for all two class combination. In both cases, prediction of new data is done by voting strategy among the build models. SVM is a computational learning process for categorising small dataset. It solves a limited quadratic optimisation problem based on structural risk minimisation to build optimum split hyperplane $f(x) = 0$ among data sets and is given by [38]:

$$y = f(x) = Z^T x + a = \sum_{i=1}^M Z_i x_i + a \quad (5)$$

where a is a scalar and Z is a M -dimensional vector. The standard dividing hyperplane is the one that creates the highest space between the plane and the nearest data, or the highest margin. The Kuhn-Tucker condition is used to convert the optimisation problem into equivalent Lagrangian dual quadratic optimisation. Further, a classification algorithm can be developed using support vectors surrounding the hyperplane.

2.3.2. K-Nearest Neighbour (KNN)

The abbreviation KNN stands for "nearest neighbour". It is a controlled machine learning algorithm and used for solving classification and regression problems. The number of closest neighbours of a new unknown variable to be predicted or classified is indicated by the symbol " K ". The main goal is to locate all the closest neighbours around a new unknown data point to find out what class it belongs to. It is a distance-based approach. KNN is an occasion-based knowledge procedure based on the idea that cases in a dataset would often be found in nearby contiguity to others along with related attributes [38]. For a given training set of classified instances $T = \{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$, where x_i is the feature vector of the unlabelled instance, y_i is the label and $y_i = c_1, c_2, \dots, c_K$; $i = 1, 2, \dots, N$. The KNN algorithm searches for the K closest examples to x based on a given distance measure for a training sample (x, y) . The vicinity including these K occurrences is characterised by $N_k(x)$. Based on the decision rules, the label of test sample x can be evaluated [38]:

$$y = \arg \max_{c_j} \sum_{x_i \in N_k(x)} IF(y_i = c_j) \quad i = 1, 2, \dots, N; j = 1, 2, \dots, K \quad (6)$$

where, IF is the indicator function.

2.3.3. Decision Tree (DT)

DT is based on learning by examples. It is a tree data structure, features of the data set can be viewed from the internal node, decision rules from the branches and results are yielded by the leaf nodes. To make the selection, the largest information acquisition feature is selected. Both have their roots in the information theory. To calculate information gain, the difference between entropy (E) before splitting and average E after splitting the dataset based on provided feature values is employed. Information gain (P, Q) relative to a collection of

examples ' P ' is determined as follows for feature ' Q ' [6]:

$$\text{Information Gain } (P, Q) = E(P) - \sum_{\mu \in \text{values}(Q)} \frac{|P_\mu|}{|P|} E(P_\mu) \quad (7)$$

where value (Q) denote the collection of all possible values for feature ' Q ', and ' P_μ ' is the sub-set of ' P ' corresponds to ' μ ' value of the feature ' Q '. The unpredictability of the set of examples (E) is measured by entropy, which is calculated by [6]:

$$E(P) = - \sum_{i=1}^n P_i \times \log_2 P_i \quad (8)$$

where ' n ' denotes the number of class, P_i is the probability of class ' i ', calculated as the ratio of the class ' i ' in the set.

2.3.4. Naive Bayes (NB)

The NB is a supervised learning algorithm that is based on the Bayesian theorem and is used to solve classification problems. It is used primarily in text classification, which includes a high-dimensional training data set. The NB classifier is one among the simplest and most effective classification algorithms to build fast machine learning models that can make fast predictions. It is a probabilistic classifier, and its prediction is based on the probability of an object. Some popular examples of NB algorithm are the filtration of spam, sentimental analysis and the classification of articles. The posterior probability of the NB classifier can be found out from the relation, and is given by [17]:

$$P(D/C) = \frac{P(C/D)P(D)}{P(C)} \quad (9)$$

where C and D denotes the characteristics of the classes and their labels. According to Bayes theorem, when all input variables C_i satisfy the conditional independent situation, the post prior probability of D in a given class n on independent inputs C_i may be stated as

$$P(D = n | C_1 \dots C_j) = \frac{\prod_{i=1}^j P(C_i | D = n)P(D = n)}{\sum_k (P(D = n_k) \prod_{i=1}^j P(C_i | D = n_i))} \quad (10)$$

The NB classifier is a widely used classifier due to its better classification performance and simplicity.

2.3.5. Random Forest (RF)

Random Forest is an artificial intelligence method which is used to determine the status of machinery components. Breiman developed the RF algorithm, which is based on decision tree construction. The training phase, bootstrapping procedure is employed to subdivide the set of features into in-bag and out-bags. For each in-bag data set, a decision tree is created, and the out-of-bag set is utilised to assess the decision tree's categorisation accuracy. Out-bag sets from the full training dataset are used to generate the algorithm's final outcomes. The majority voting scheme is adopted to yield the final classifier accuracy [39].

3. Descriptions of experimental dataset

The proposed approach is validated on three datasets, including one self-collected datasets, and two open-source datasets. The proposed approach is applied to only localised fault conditions on the system.

3.1. Self-collected dataset

Self-Collected dataset is denoted by Dataset-1 in this work. It is acquired from the experimental test setup installed in Dynamics of Machine Laboratory, IIT Patna. Fig. 2 shows the test setup which comprises of bevel gearbox, belt-pulley drive, shaft, and motor. Table 1 shows the specification of the test setup and important characteristic frequency calculation for gearbox. Triaxial accelerometer are mounted on the bevel gearbox for vibration data acquisition and data along three axes are used in this work. In the experiment, two localised

Table 1
Specifications of test setup.

Parameters	Values
Gear tooth, n_g	27
Pinion tooth, n_p	18
Pulley diameter of pinion shaft, d_p	0.125 m
Pulley diameter of motor shaft, d_m	0.05 m
Frequency of rotating pinion, f_p	$\frac{d_m}{d_p} \times \text{Motor speed (Hz)}$
Frequency of rotating gear, f_g	$\frac{n_p}{n_g} \times f_p$ (Hz)
Gear Mesh Frequency (GMF), f_m	$f_p \times n_p$ Or, $f_g \times n_g$ (Hz)
GMF Harmonics calculation	GMF 1 = $1 \times f_m$, GMF 2 = $2 \times f_m$ and so on

Table 2
Description for Dataset-1 (Bevel gearbox).

Fault conditions	Healthy tooth			Missing tooth			Chipped tooth		
Motor speed (Hz)	15	25	35	15	25	35	15	25	35
Pinion speed (Hz)	6	10	14	6	10	14	6	10	14
Load (lb-ft)	0, 2, 4	0, 2, 4	0, 2, 4	0, 2, 4	0, 2, 4	0, 2, 4	0, 2, 4	0, 2, 4	0, 2, 4
Sampling rate (Hz)	2048								
Recorded time (s)	30								
Sensor	A tri-axial accelerometer is used.								
Total samples collected	27			27			27		

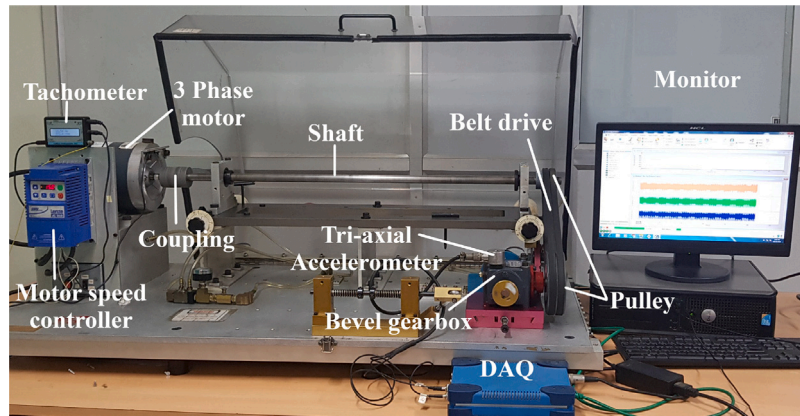


Fig. 2. Experimental set-up for Dataset-1.



Fig. 3. Localised faults condition for Dataset-1.

fault conditions on bevel gear teeth, namely missing and chipped with healthy conditions, were used to collect samples. Fig. 3 shows these fault conditions. Total 81 samples were recorded. The other parameters related to data acquisition are given in Table 2.

3.2. Open-source dataset

Two open-source dataset namely PHM2009 dataset and XJTUSpurgear dataset are used in this work in which PHM2009 dataset

is denoted by Dataset-2, and XJTUSpurgear dataset is denoted by Dataset-3

3.2.1. Dataset-2

PHM2009 dataset were provided by the Prognostics and Health Management (PHM) Society in 2009 data challenge competition. The brief description about the specification of experimental test set-up and data collection procedure are given in [2,40,41]. Two gearbox datasets such as spur and helical datasets were provided under different label

Table 3

Description for Dataset-2 (Helical gearbox).

Fault conditions	Label1-Healthy		Label2-Chipped		Label5-Broken+BIF	
Motor speed (Speed step) (Hz)	30–50(5)		30–50(5)		30–50(5)	
Load (lb-ft)	High	Low	High	Low	High	Low
No. of Trails	2	2	2	2	2	2
Sampling rate (KHz)	66.67					
Recorded time (s)	3					
Sensor	Accelerometer data pertaining to Column 2 (output end) is used.					
Total samples collected	20		20		20	

Table 4

Description for Dataset-3 (Spur gearbox).

Fault conditions	Healthy	Crack-0.6 mm	Crack-1.0 mm	Crack-1.4 mm
Motor speed (rpm)	900, 1200, 0–1200–0 (Varying speed)			
Sampling rate (KHz)	10			
Recorded time (s)	27			
Sensor	Accelerometer sensor data collected from 12 different locations is used.			
Total samples collected	36	36	36	36

(carries localised or both localised and distributed faults). Since our work is restricted to only localised faults, therefore only labels containing localised faults of helical gearbox are considered for analysis. Total 60 samples were taken in this work. The other parameters related to this dataset are given in Table 3.

3.2.2. Dataset-3

XJTUSpurgear dataset are provided by the Institute of Aero-engine at Xi'an Jiaotong University (XJTU), Shaanxi, P.R. China that allow anyone to use and verify the diagnosis problem [42]. The details description about the experimental setup and data acquisition are given in literature [42]. Three types of root cracks on spur gear with crack depths of 0.6 mm, 1 mm, and 1.4 mm with healthy tooth are analysed at constant and varying speeds in this work. Total 144 samples were taken for analysis. The details of this dataset used in this work are given in Table 4.

4. Results and discussion

In this section, some key conclusion about LHOM trend under different localised fault conditions is discussed in Section 4.1, types of features extracted in Section 4.2, selection of dominant features through different feature selection technique in Section 4.3, and accuracy comparison with five ML-classifiers in Section 4.4.

4.1. Behaviour of LHOM under localised fault conditions

To understand the behaviour or trend of LHOM with respect to order of moment and severity of localised faults on the different gearboxes under different operating conditions, LHOM features up to 100 order i.e., $k = [1, 2, 3, \dots, 100]$ are extracted from each dataset by using the Eq. (3), respectively. Figs. 4–7 depict the variation of LHOM features with order of moment (k) for Dataset-1, Dataset-2, and Dataset-3, respectively. These figures also contain the zoom-view up to order 10 to show how LHOM features behave under different localised fault conditions with initial order of moments.

The following observations may be taken from these figures:

- At the initial stage up to the 5th order of moment, the LHOM shows a mixed trend with order of moment at each fault condition. While, beyond the 5th order moment, the LHOM shows a linear trend with order of moment for each dataset, respectively.
- In all three datasets, the amplitude of LHOM (beyond 5th order) is directly proportional to the severity of faults in each dataset such as in Dataset-1, at a particular order, the LHOM value of Missing

> Chipped > Healthy (see Fig. 4). Similarly, for Dataset-2, the LHOM value of Broken+BIF > Chipped > Healthy (see Fig. 5), and for Dataset-3, the LHOM value of Crack 1.4 mm > Crack 1 mm > Crack 0.6 mm > Healthy (see Figs. 6–7).

- Also, this trend is directly proportional to the order of moments i.e., the LHOM values increases with the severity of fault and order of moment.
- The demarcation range among the LHOM to categorise the faults is getting more clear with the increase of order of moment.
- The above results are generalised and checked with different gearboxes under different localised fault conditions at different operating parameters like constant speed (see Figs. 4–6); varying speed (see Fig. 7), different load conditions like no load (see Fig. 4); high load (see Fig. 5).

In conclusion, LHOM features can be used to distinguish between various localised faults in the gearbox. Each of these features has its own significance in predicting the severity of the faults in the system and it plays an essential part in the fault categorisation of gearboxes, especially in localised faults.

4.2. Feature extraction

Two types of features: (1) conventional time domain features; and (2) proposed LHOM features are extracted from each dataset for comparative analysis. Table 5 shows the formulation of 10 widely used conventional time domain features (with the maximum 4th order moment used) [8–11,14]. For effective comparison with conventional one, the same number of proposed LHOM features i.e., up to order 10 are calculated by using Eq. (3).

4.3. Feature selection

The selection of features is one of the most essential components of fault diagnostics. The performance of the classifier is impacted by the excess amount and quality of features, which also contribute to the over-fitting outcome. In order to choose the best feature subsets for the ML algorithm, the selection of features is an essential step. In this work, three different feature selection techniques i.e., PCC, NCA, and RF are used to select contributing features from the both types of features (proposed and conventional) as discussed in above subsection. As the total number of features is 10 for both type, the number of selected features are varied from 2 to 10, and each selected feature set is fed to the ML-classifier to find the optimal set of features.

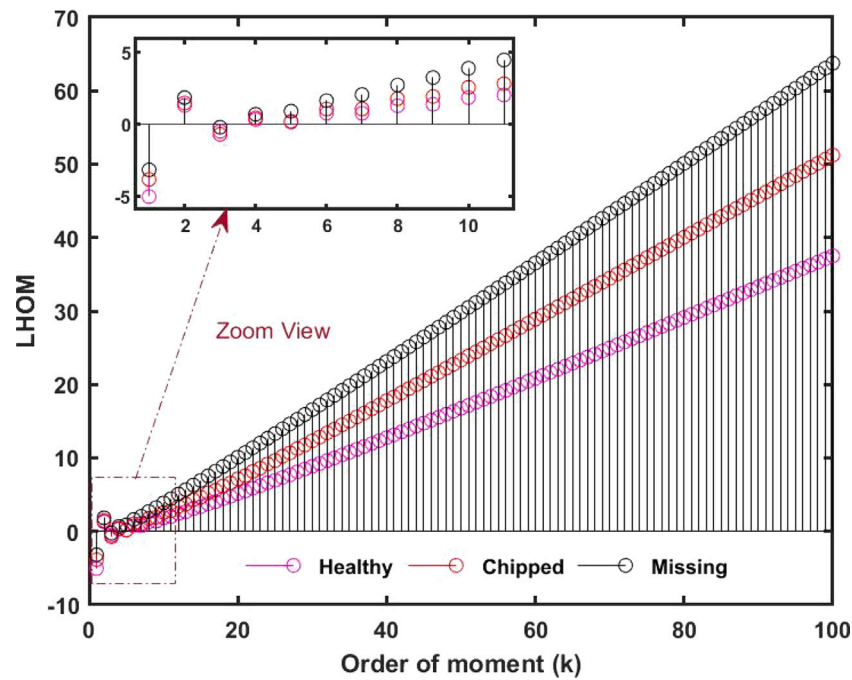


Fig. 4. LHM variation for Dataset-1 at 25 Hz and 0 lb-ft load condition.

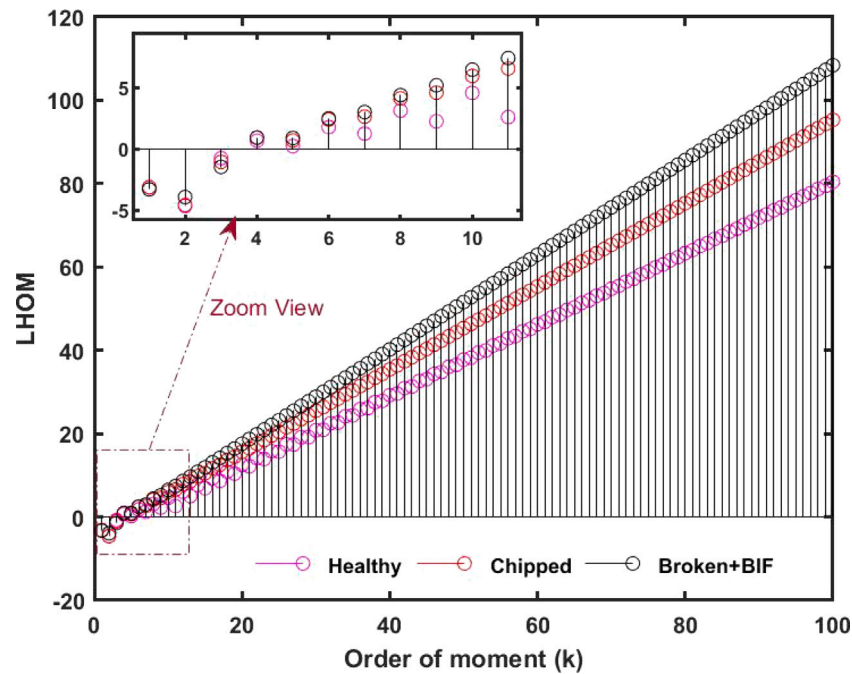


Fig. 5. LHM variation for Dataset-2 at 40 Hz and high load condition.

4.4. Fault classification

In this work, five ML-classifiers such as KNN, MCSVM, DT, RF, and NB are combined with the three feature selection technique i.e., PCC, NCA, and RF to demonstrate the effectiveness of both type of extracted features (conventional and proposed) in gearbox fault classification. The optimal features set obtained from each feature selection technique are input to these ML-classifiers one by one for each dataset. The model developed from each classifier is trained and tested by the cross validation (K-fold) technique, which varies from 2 to 9 fold cross validation. Also, based on the fitness function (maximum classification accuracy),

the key parameters of ML-classifiers, i.e., distance type, number of neighbours in KNN, kernel function in MCSVM, number of splits in DT, number of trees in RF, and predictor distribution function (PDF) in NB are optimised. In KNN, a total of 10 distance types (details are given in [43,44]) such as Euclidean, Seclidean, Cosine, Correlation, City-block, Chebychev, Hamming, Jaccard, Minkowski, and Spearman, with a number of neighbours varying from 1 to 10, are used to optimise the classification accuracy. Similarly, for MCSVM, three kernel functions such as radial basis function (rbf), gaussian, and linear; and for NB, three PDFs such as normal (Gaussian), kernel, and multivariate multinomial are used to optimise the classification accuracy of the respective

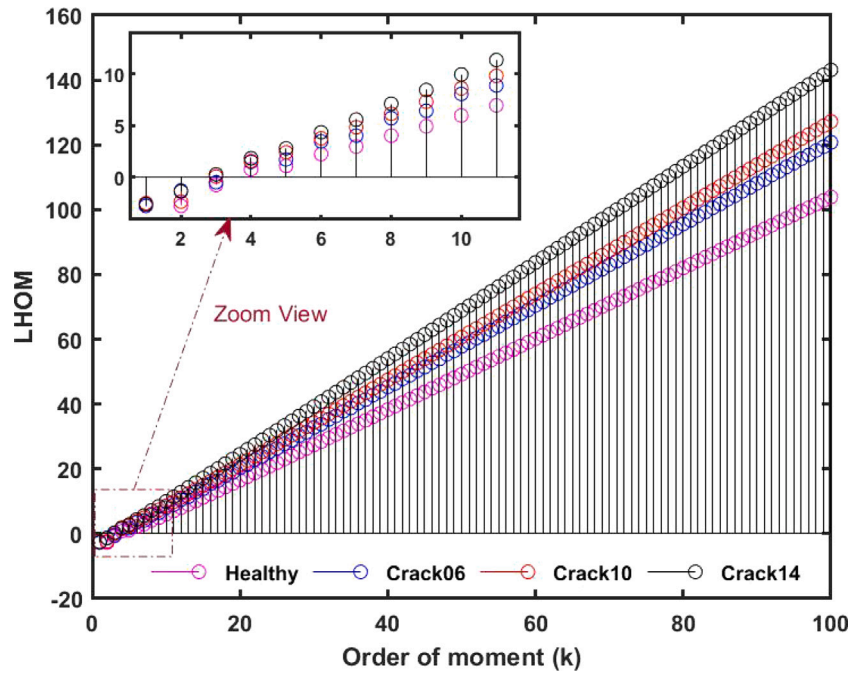


Fig. 6. LHM variation for Dataset-3 at 900 rpm condition.

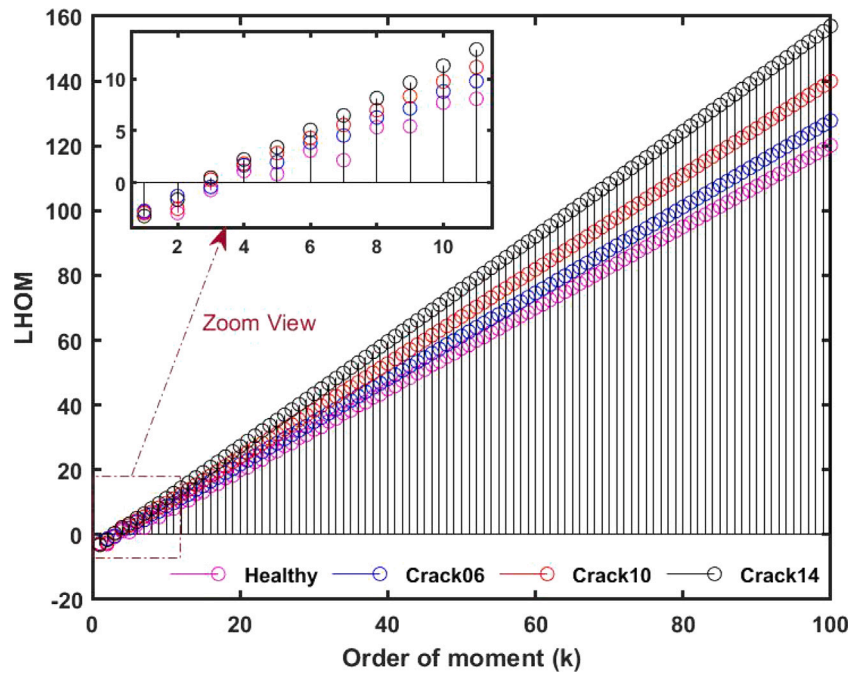


Fig. 7. LHM variation for Dataset-3 at 0-1200-0 rpm (varying speed) condition.

classifier. The details of these functions are given in [44]. Tables 6–8 show the optimised number of selected features, optimised K-fold, optimised parameters and optimised accuracy of the five ML-classifiers at PCC, NCA, and RF with Dataset-1, Dataset-2, and Dataset-3 for both proposed and conventional approach, respectively. Tables clearly show that at some places (marked with bold) the proposed approach is dominating, at some places the conventional approach is dominating, and

at some places both approaches are sharing the same accuracy. Fig. 8 shows a comprehensive overview of both types of features (proposed and conventional) with respect to the number of instances (Total = 3(Datasets) $\times$ 3(Feature selection techniques) $\times$ 5(ML-Classifiers) = 45), i.e., the bar chart shows at how many instances both types of features are superior to each other and at how many instances they share the same accuracy. Fig. 8(a) shows, with respect to ML-classifiers, that for

Table 5
Time domain features based on up to 4th order moment [8–11,14].

Feature	Expression	Feature	Expression
Peak	$\max Y(n) $	Std. deviation (σ)	$\sqrt{\frac{\sum_{n=1}^N (Y(n)-\mu)^2}{N-1}}$
Mean (μ)	$\frac{\sum_{n=1}^N Y(n)}{N}$	Skewness	$\frac{\sum_{n=1}^N (Y(n)-\mu)^3}{(N-1)\sigma^3}$
RMS	$\sqrt{\frac{\sum_{n=1}^N Y(n)^2}{N}}$	Kurtosis	$\frac{\sum_{n=1}^N (Y(n)-\mu)^4}{(N-1)\sigma^4}$
Crest factor	$\frac{Peak}{RMS}$	Shape factor	$\frac{RMS}{\frac{1}{N} \sum_{n=1}^N Y(n) }$
Impulse factor	$\frac{Peak}{\frac{1}{N} \sum_{n=1}^N Y(n) }$	Clearance factor	$\frac{Peak}{\left(\frac{1}{N} \sum_{n=1}^N \sqrt{ Y(n) }\right)^2}$

Where N is the total number of data points in the input signal Y .

Table 6
Comparison table with PCC feature selection technique.

Dataset	ML classifiers	10 Time domain features				10 LHOM features			
		OSF	OPs	OKfold	OAccuracy (%)	OSF	OPs	OKfold	OAccuracy (%)
Dataset-1	MCSVM	3	KF=Linear	9	91.36	9	KF=Linear	3	90.12
	KNN	3	Dt=Seuclidean, NN=5	8	92.60	8	Dt=Chebychev, NN=4	4	91.36
	DT	2	Ns=20	7	91.36	8	Ns=20	9	88.89
	RF	5	Nt=20	3	92.60	6	Nt=20	6	91.36
	NB	5	PDF=Normal	7	91.36	8	PDF=Kernel	3	88.89
Dataset-2	MCSVM	10	KF=Linear	9	81.67	2	KF=rbf	7	98.33
	KNN	10	Dt=Seuclidean, NN=1	7	100.00	2	Dt=Euclidean, NN=1	7	100.00
	DT	9	Ns=20	4	88.33	4	Ns=20	4	85.00
	RF	4	Nt=20	7	90.08	7	Nt=20	9	93.12
	NB	10	PDF=Kernel	3	83.33	4	PDF=Normal	6	85.00
Dataset-3	MCSVM	3	KF=Linear	10	79.86	9	KF=rbf	9	84.03
	KNN	5	Dt=Cosine, NN=4	6	84.72	9	Dt=Cosine, NN=1	4	81.25
	DT	9	Ns=20	7	81.94	9	Ns=20	9	75.70
	RF	10	Nt=20	8	81.94	10	Nt=20	7	80.60
	NB	8	PDF=Kernel	4	74.31	9	PDF=Kernel	9	74.31

OSF=Optimised selected features, OPs=Optimised parameters, OKfold=Optimised Kfold, OAccuracy=Optimised Accuracy, NN= Number of neighbours, Dt= Distance type, KF= Kernel function, Ns= Number of splits, Nt= Number of tree, PDF= Predictor distribution function

MCSVM, 8/9 instances are superior to the conventional one, and 1/9 shows a marginal difference in accuracy (see Table 6), whereas for RF, 7/9 instances are superior to the proposed one. Rest classifiers show contradictory results for both approaches. Fig. 8(b) shows, with respect to feature selection technique, it is clearly observed that the NCA and RF are more superior technique with proposed one. Fig. 8(c) shows, with respect to datasets, Dataset-1 shows shared trend, Dataset-2 shows superior than conventional one, and Dataset-3 shows mixed trend. In all the subfigures of Fig. 8, out of 45 instances, 21 instances with proposed one are dominant, 20 instances with conventional one are dominant, and 4 instances are shared, respectively. Overall, by considering the major possibilities in terms of different datasets, different fault situations, different feature selection techniques, and different well-known ML-classifiers, the proposed one is marginally superior to the conventional one. However, the proposed LHOM features with the MCSVM classifier outperformed the rest of the classifiers and demonstrated a significant improvement in classification accuracy at PCC, NCA, and RF feature selection techniques (see Table 9). The average increase in accuracy with the MCSVM classifier at PCC is 8.09%, at NCA it is 4.69%, and at RF it is 10.11%.

In conclusion, the proposed LHOM features with the MCSVM classifier perform well under the different major possibilities taken in this work. The obtained results clearly state that the proposed LHOM features can also be used as an extracted feature for diagnosing localised gearbox faults.

5. Conclusion

In this research work, new features based on the logarithmic amplitude of HOM (LHOM) were proposed for diagnosing localised gearbox faults. The LHOM features are unique and are being applied for the first time for the fault diagnostics of the gearbox. Three benchmark datasets taken from different gearboxes (such as spur, helical, and bevel) at different localised fault conditions were taken in this work for result validation. The results showed that the LHOM features had a linear relationship with the order of moments and were directly proportional to the severity of faults beyond the fifth order moment. With increasing order of moments, the demarcation range for categorising different depths of localised faults becomes more clear. Further, based on different case studies on both types of feature approaches

Table 7
Comparison table with NCA feature selection technique.

Dataset	ML classifiers	10 Time domain features				10 LHOM features			
		OSF	OPs	OKfold	OAccuracy (%)	OSF	OPs	OKfold	OAccuracy (%)
Dataset-1	MCSVM	2	KF=Linear	5	88.89	3	KF=rbf	6	91.36
	KNN	2	Dt=Seuclidean, NN=6	5	91.36	2	Dt=Chebychev, NN=6	9	93.83
	DT	2	Ns=20	2	86.42	3	Ns=20	9	90.12
	RF	10	Nt=20	3	91.36	8	Nt=20	5	92.60
	NB	3	PDF=Normal	5	90.12	3	PDF=Kernel	7	90.12
Dataset-2	MCSVM	3	KF=Linear	7	81.67	5	KF=rbf	8	85.00
	KNN	5	Dt=Seuclidean, NN=1	4	100.00	8	Dt=Seuclidean, NN=1	5	93.33
	DT	7	Ns=20	4	83.33	7	Ns=20	4	85.00
	RF	10	Nt=20	4	93.33	9	Nt=20	8	92.19
	NB	8	PDF=Kernel	7	85.00	10	PDF=Normal	6	83.33
Dataset-3	MCSVM	5	KF=rbf	3	77.08	5	KF=rbf	9	82.64
	KNN	5	Dt=Cosine, NN=8	3	84.72	5	Dt=Seuclidean, NN=1	5	82.64
	DT	10	Ns=20	9	81.25	6	Ns=20	10	77.78
	RF	10	Nt=20	9	82.64	5	Nt=20	5	82.00
	NB	6	PDF=Normal	3	75.00	2	PDF=Normal	3	77.78

Table 8
Comparison table with RF feature selection technique.

Dataset	ML classifiers	10 Time domain features				10 LHOM features			
		OSF	OPs	OKfold	OAccuracy (%)	OSF	OPs	OKfold	OAccuracy (%)
Dataset-1	MCSVM	2	KF=Linear	5	88.89	5	KF=Linear	5	91.36
	KNN	2	Dt=Seuclidean, NN=6	5	91.36	5	Dt=Seuclidean, NN=6	8	93.83
	DT	3	Ns=20	6	87.65	6	Ns=20	9	91.36
	RF	6	Nt=20	6	92.60	5	Nt=20	6	91.36
	NB	5	PDF=Normal	7	91.36	6	PDF=Normal	5	90.12
Dataset-2	MCSVM	2	KF=Linear	9	81.67	2	KF=rbf	7	98.33
	KNN	9	Dt=Seuclidean, NN=1	7	100.00	2	Dt=Euclidean, NN=1	7	100.00
	DT	4	Ns=20	4	86.67	6	Ns=20	4	88.33
	RF	3	Nt=20	7	98.41	3	Nt=20	4	91.67
	NB	10	PDF=Kernel	3	83.33	4	PDF=Normal	6	85.00
Dataset-3	MCSVM	8	KF=Linear	4	77.78	8	KF=rbf	9	83.33
	KNN	10	Dt=Correlation, NN=2	3	84.72	6	Dt=Cosine, NN=6	5	86.11
	DT	10	Ns=20	5	81.25	3	Ns=20	9	76.39
	RF	10	Nt=20	7	82.59	6	Nt=20	5	80.56
	NB	7	PDF=Kernel	4	74.31	4	PDF=Kernel	5	76.40

Table 9
% Change in accuracy of LHOM approach over time domain approach using MCSVM classifier.

S.No.	Feature selection technique	% Increase in optimised accuracy			Average (%)
		Dataset-1	Dataset-2	Dataset-3	
1	PCC	−1.36	20.40	5.22	8.09
2	NCA	2.78	4.07	7.21	4.69
3	RF	2.78	20.40	7.13	10.11

(proposed and conventional), it was found that the MCSVM classifier with proposed features outperformed the rest of the classifiers and showed significant improvements in classification accuracy at PCC, NCA, and RF feature selection techniques. Overall, the proposed LHOM feature show marginal increment for all well known ML-classifiers and show significant improvement for MCSVM classifier. The outcomes show that the proposed LHOM feature can be used as a substitute for the conventional one. In the future, the findings of these studies will help to provide important guidelines for understanding the behaviour of LHOM in detecting faults in other machinery components as well.

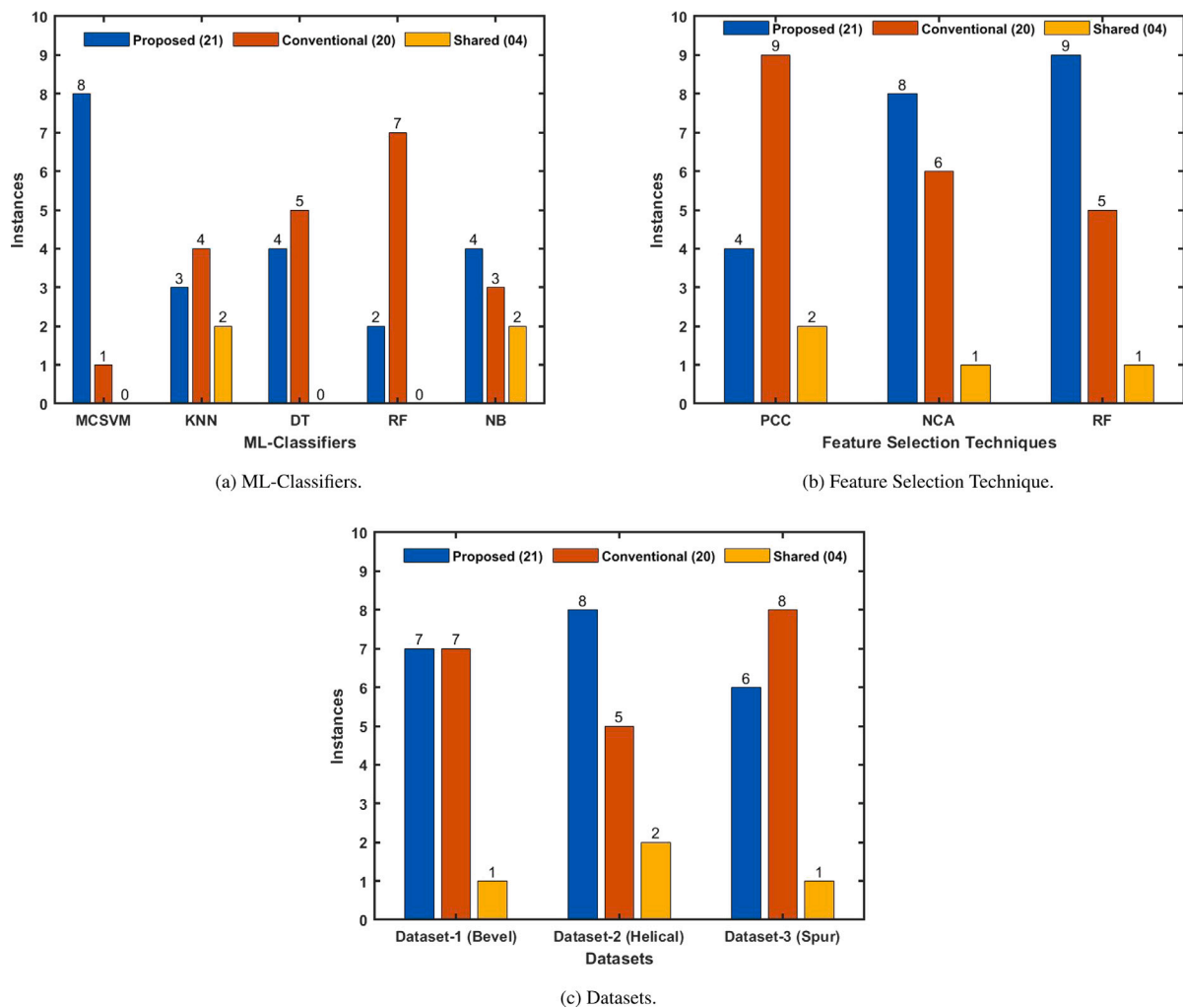


Fig. 8. Bar chart of accuracy comparison with respect to number of instances (Total=45).

CRedit authorship contribution statement

Sanjeev Kumar: Conceptualization, Software, Data curation, Visualization, Formal analysis, Writing – original draft. **Vikash Kumar:** Conceptualization, Methodology, Software, Formal analysis, Validation, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Somnath Sarangi:** Supervision, Resources, Conceptualization, Writing – review & editing. **Om Prakash Singh:** Data curation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

This research was supported by Dynamics of Machines Laboratory, Indian Institute of Technology Patna, India.

References

- [1] A. Parey, A. Singh, Gearbox fault diagnosis using acoustic signals, continuous wavelet transform and adaptive neuro-fuzzy inference system, *Appl. Acoust.* 147 (2019) 133–140.
- [2] T. Han, C. Liu, W. Yang, D. Jiang, Learning transferable features in deep convolutional neural networks for diagnosing unseen machine conditions, *ISA Trans.* 93 (2019) 341–353.
- [3] M. Cerrada, G. Zurita, D. Cabrera, R.-V. Sánchez, M. Artés, C. Li, Fault diagnosis in spur gears based on genetic algorithm and random forest, *Mech. Syst. Signal Process.* 70 (2016) 87–103.
- [4] C. Li, R.-V. Sanchez, G. Zurita, M. Cerrada, D. Cabrera, R.E. Vásquez, Gearbox fault diagnosis based on deep random forest fusion of acoustic and vibratory signals, *Mech. Syst. Signal Process.* 76 (2016) 283–293.
- [5] S. Mukherjee, R. Kaushal, V. Kumar, S. Sarangi, A novel approach of gearbox fault diagnosis by using time synchronous averaging and J48 algorithm, in: *Advances in Electromechanical Technologies*, Springer, 2021, pp. 927–935.
- [6] V. Kumar, A. Verma, S. Sarangi, Fault diagnosis of single-stage bevel gearbox by energy operator and J48 algorithm, in: *Advances in Condition Monitoring and Structural Health Monitoring*, Springer, 2021, pp. 231–239.
- [7] Y. Zhang, W. Lu, F. Chu, Planet gear fault localization for wind turbine gearbox using acoustic emission signals, *Renew. Energy* 109 (2017) 449–460.
- [8] Vanraj, S. Dhami, B. Pabla, Hybrid data fusion approach for fault diagnosis of fixed-axis gearbox, *Struct. Health Monit.* 17 (4) (2018) 936–945.
- [9] P.H. Rodriguez, J.B. Alonso, M.A. Ferrer, C.M. Travieso, Application of the Teager–Kaiser energy operator in bearing fault diagnosis, *ISA Trans.* 52 (2) (2013) 278–284.
- [10] C. Shen, D. Wang, F. Kong, W.T. Peter, Fault diagnosis of rotating machinery based on the statistical parameters of wavelet packet paving and a generic support vector regressive classifier, *Measurement* 46 (4) (2013) 1551–1564.

- [11] D. Han, N. Zhao, P. Shi, A new fault diagnosis method based on deep belief network and support vector machine with Teager–Kaiser energy operator for bearings, *Adv. Mech. Eng.* 9 (12) (2017) 1687814017743113.
- [12] T. Buckley, B. Ghosh, V. Pakrashi, A feature extraction & selection benchmark for structural health monitoring, *Struct. Health Monit.* (2022) 14759217221111141.
- [13] A. Widodo, B.-S. Yang, Support vector machine in machine condition monitoring and fault diagnosis, *Mech. Syst. Signal Process.* 21 (6) (2007) 2560–2574.
- [14] V. Kumar, S. Mukherjee, A.K. Verma, S. Sarangi, An AI-based nonparametric filter approach for gearbox fault diagnosis, *IEEE Trans. Instrum. Meas.* 71 (2022) 1–11, <http://dx.doi.org/10.1109/TIM.2022.3186700>.
- [15] D. Wang, K-nearest neighbors based methods for identification of different gear crack levels under different motor speeds and loads: Revisited, *Mech. Syst. Signal Process.* 70 (2016) 201–208.
- [16] X. Tang, X. Gu, L. Rao, J. Lu, A single fault detection method of gearbox based on random forest hybrid classifier and improved Dempster–Shafer information fusion, *Comput. Electr. Eng.* 92 (2021) 107101.
- [17] J. Singh, M. Azamfar, F. Li, J. Lee, A systematic review of machine learning algorithms for PHM of rolling element bearings: fundamentals, concepts, and applications, *Meas. Sci. Technol.* (2020).
- [18] P. Cao, S. Zhang, J. Tang, Preprocessing-free gear fault diagnosis using small datasets with deep convolutional neural network-based transfer learning, *IEEE Access* 6 (2018) 26241–26253.
- [19] Z. Chen, C. Li, R.-V. Sanchez, Gearbox fault identification and classification with convolutional neural networks, *Shock Vib.* 2015 (2015).
- [20] C.D. Nguyen, A. Prosvirni, J.-M. Kim, A reliable fault diagnosis method for a gearbox system with varying rotational speeds, *Sensors* 20 (11) (2020) 3105.
- [21] P. Srihari, K. Govindarajulu, K. Ramachandra, A method to improve reliability of gearbox fault detection with artificial neural networks, *Int. J. Automot. Mech. Eng.* 2 (2010) 221–230.
- [22] F. Honarvar, H. Martin, New statistical moments for diagnostics of rolling element bearings, *J. Manuf. Sci. Eng.* 119 (3) (1997) 425–432, <http://dx.doi.org/10.1115/1.2831123>.
- [23] V. Kumar, A. Rai, S. Mukherjee, S. Sarangi, A Lagrangian approach for the electromechanical model of single-stage spur gear with tooth root cracks, *Eng. Fail. Anal.* 129 (2021) 105662.
- [24] V. Kumar, A. Kumar, S. Kumar, S. Sarangi, TVMS calculation and dynamic analysis of carburized spur gear pair, *Mech. Syst. Signal Process.* 166 (2022) 108436.
- [25] S.R. Saufi, Z.A.B. Ahmad, M.S. Leong, M.H. Lim, Gearbox fault diagnosis using a deep learning model with limited data sample, *IEEE Trans. Ind. Inform.* 16 (10) (2020) 6263–6271.
- [26] J. Obuchowski, R. Zimroz, A. Wylomańska, Blind equalization using combined skewness–kurtosis criterion for gearbox vibration enhancement, *Measurement* 88 (2016) 34–44.
- [27] H. Martin, Statistical moment analysis as a means of surface damage detection, in: *Proceedings of the 7th International Modal Analysis Conference*, Vol. 1, Society for Experimental Mechanics Las Vegas, USA, 1989, pp. 1016–1021.
- [28] A.S. Sait, Y.I. Sharaf-Eldeen, A review of gearbox condition monitoring based on vibration analysis techniques diagnostics and prognostics, in: *Rotating Machinery, Structural Health Monitoring, Shock and Vibration*, Volume 5, Springer, 2011, pp. 307–324.
- [29] P.D. Samuel, D.J. Pines, A review of vibration-based techniques for helicopter transmission diagnostics, *J. Sound Vib.* 282 (1–2) (2005) 475–508.
- [30] H.J. Decker, Spiral Bevel Pinion Crack Detection in a Helicopter Gearbox, National aeronautics and space administration cleveland oh glenn research center, 2003.
- [31] J. Wang, Y. Ma, Z. Huang, R. Xue, R. Zhao, Performance analysis and enhancement of deep convolutional neural network, *Bus. Inf. Syst. Eng.* 61 (3) (2019) 311–326.
- [32] E.S. Glos, Higher-order statistics applications in image sequence processing (Ph.D. thesis), Universitat Politècnica de Catalunya (UPC), 1995.
- [33] X. Niu, L. Zhu, H. Ding, New statistical moments for the detection of defects in rolling element bearings, *Int. J. Adv. Manuf. Technol.* 26 (11–12) (2005) 1268–1274.
- [34] B. Mwangi, T.S. Tian, J.C. Soares, A review of feature reduction techniques in neuroimaging, *Neuroinformatics* 12 (2) (2014) 229–244.
- [35] W. Yang, K. Wang, W. Zuo, Neighborhood component feature selection for high-dimensional data, *J. Comput.* 7 (1) (2012) 161–168.
- [36] M. Robnik-Šikonja, I. Kononenko, Theoretical and empirical analysis of ReliefF and RReliefF, *Mach. Learn.* 53 (1) (2003) 23–69.
- [37] K. Kira, L.A. Rendell, A practical approach to feature selection, in: *Machine Learning Proceedings 1992*, Elsevier, 1992, pp. 249–256.
- [38] R. Liu, B. Yang, E. Zio, X. Chen, Artificial intelligence for fault diagnosis of rotating machinery: A review, *Mech. Syst. Signal Process.* 108 (2018) 33–47.
- [39] V. Vakharia, V. Gupta, P. Kankar, Efficient fault diagnosis of ball bearing using ReliefF and random forest classifier, *J. Braz. Soc. Mech. Sci. Eng.* 39 (8) (2017) 2969–2982.
- [40] X. Li, Y. Hu, M. Li, J. Zheng, Fault diagnostics between different type of components: A transfer learning approach, *Appl. Soft Comput.* 86 (2020) 105950.
- [41] H. Al-Atat, D. Siegel, J. Lee, A systematic methodology for gearbox health assessment and fault classification, *Int. J. Progn. Health Manag. Soc.* 2 (1) (2011) 16.
- [42] T. Li, Z. Zhou, S. Li, C. Sun, R. Yan, X. Chen, The emerging graph neural networks for intelligent fault diagnostics and prognostics: A guideline and a benchmark study, *Mech. Syst. Signal Process.* 168 (2022) 108653.
- [43] K. Chomboon, P. Chujai, P. Teerarassamee, K. Kerdprasop, N. Kerdprasop, An empirical study of distance metrics for k-nearest neighbor algorithm, in: *Proceedings of the 3rd International Conference on Industrial Application Engineering*, 2015, pp. 280–285.
- [44] T. Hastie, R. Tibshirani, J.H. Friedman, J.H. Friedman, *The elements of statistical learning: data mining, inference, and prediction*, Vol. 2, Springer, 2009.